

PHY110

ENGINEERING PHYSICS

Lecture #0

The kick start session



L OVELY
P ROFESSIONAL
U NIVERSITY

Transforming Education Transforming India

University Vision and Mission



**To be a premier academic institution,
recognized internationally for its
contribution to industry and society
through excellence in teaching,
learning, research, internationalization,
entrepreneurship and leadership.**



- To transform education through academic rigor, practical orientation and outcome-based teaching.
- To develop and implement a relationship of cooperation between industry and academia.
- To undertake impactful research addressing local, national and global challenges.
- To prepare graduates to be lifelong learners with strong analytical and leadership skills.
- To develop global professionals and entrepreneurs with innovative spirit, tolerance and desire to make a difference to the society.

Programme Outcomes

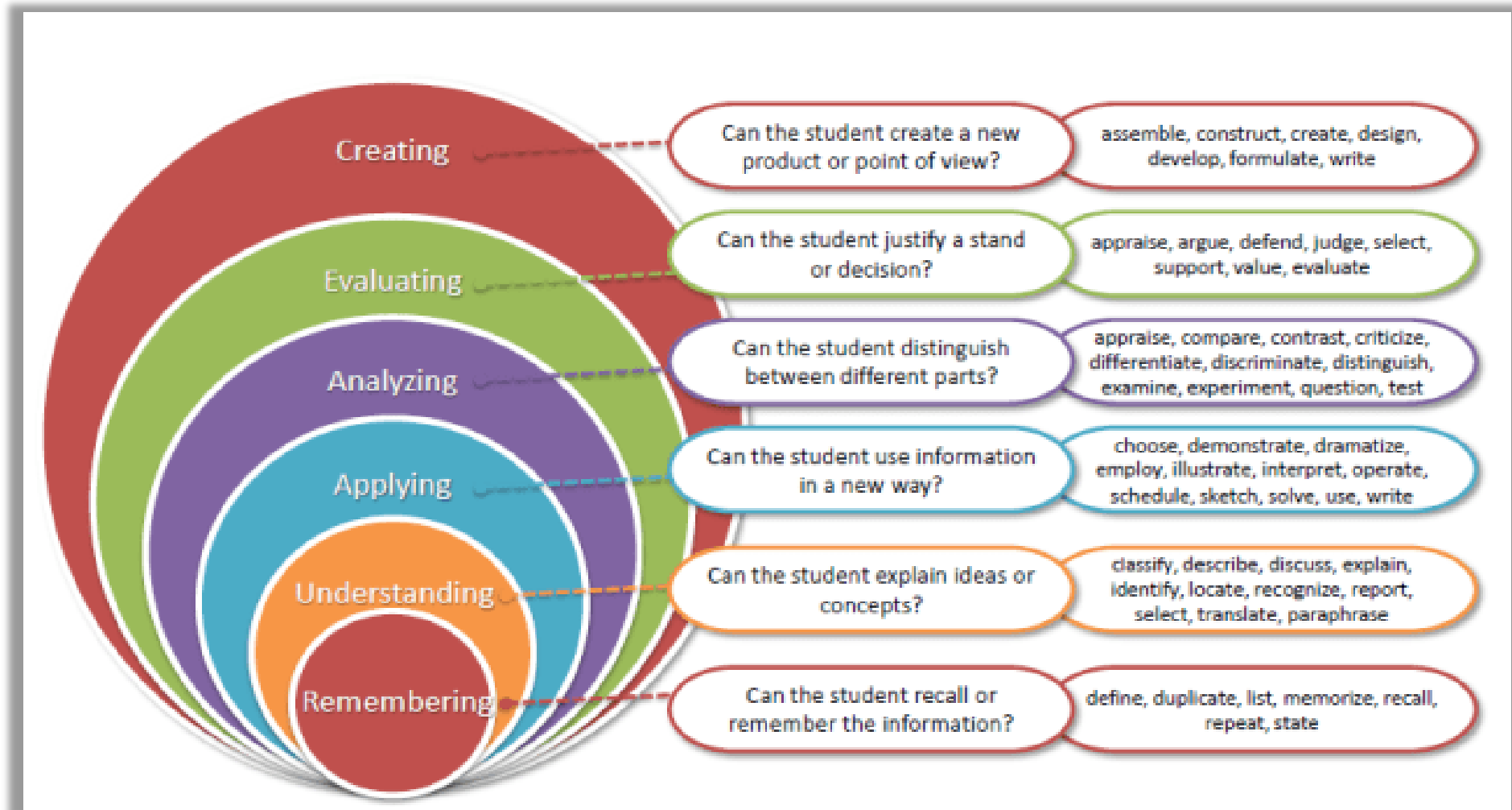
- PO1 : **Engineering knowledge** : Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- PO2 : **Problem analysis** : Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- PO3 : **Design/development of solutions** : Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- PO4 : **Conduct investigations of complex problems** : Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- PO5 : **Modern tool usage** : Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations
- PO6 : **The engineer and society** : Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- PO7 : **Environment and sustainability** : Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

- PO8 : **Ethics** : Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- PO9 : **Individual and team work** : Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- PO10 : **Communication** : Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- PO11 : **Project management and finance** : Demonstrate knowledge and understanding of the engineering, management principles and apply the same to one's own work, as a member or a leader in a team, manage projects efficiently in respective disciplines and multidisciplinary environments after consideration of economic and financial factors.
- PO12 : **Life-long learning** : Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Programme Specific Outcomes

- PSO1 : Apply basic knowledge in the areas such as Software Engineering, Networking and Security, Database Management Systems, Intelligent Systems, Operating Systems and System Architecture for building Software products.
- PSO2 : Attain the ability to design and develop software based systems, evaluate and recognize potential risks and provide creative, economical and optimal solutions.
- PSO3 : Provide effective and efficient real time solutions using attained knowledge in inter-disciplinary domains for societal benefits through projects.
- PSO4 : Ability to adapt future technologies through acquired skills and knowledge to ideate, design, test and deploy solutions for different business environments.

Revised Bloom's Taxonomy



Course outcomes

Course Outcomes: Through this course students should be able to

CO1 :: explore the basic principles of physics for the foundation for various engineering courses.

CO2 :: understand the principle and working of lasers and optical fibres for their wide applications.

CO3 :: illustrate the development of quantum mechanics and its applications.

CO4 :: articulate the physics of solids to understand their properties.

CO5 :: associate the knowledge of engineering materials with their applications.



What you will study here in this course?

Course Content



But how to find the syllabus in UMS ?

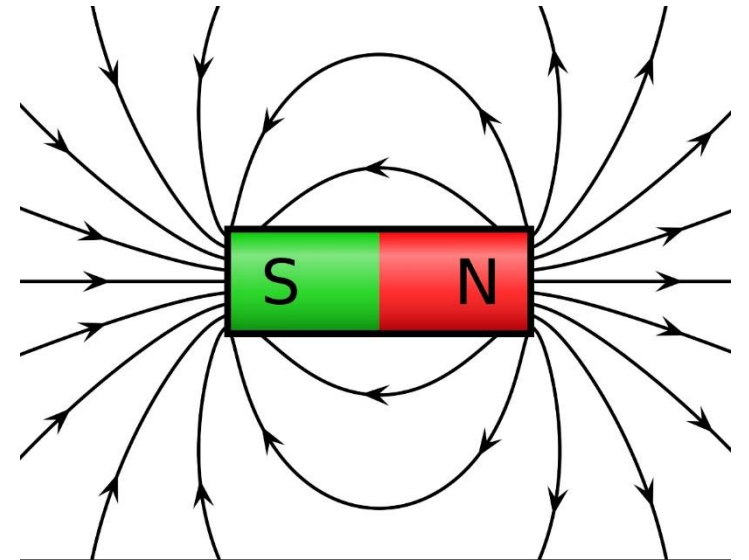
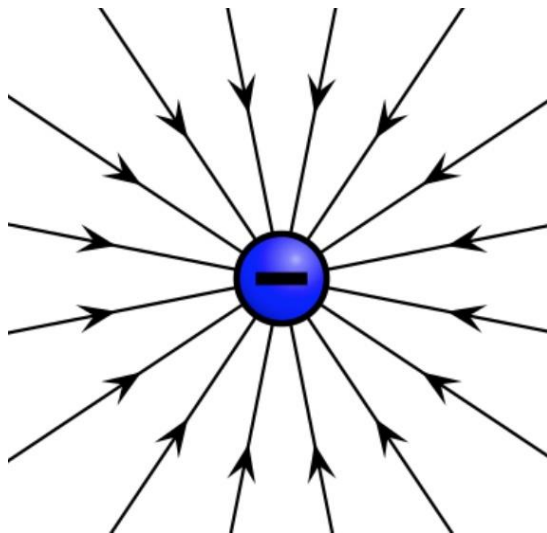
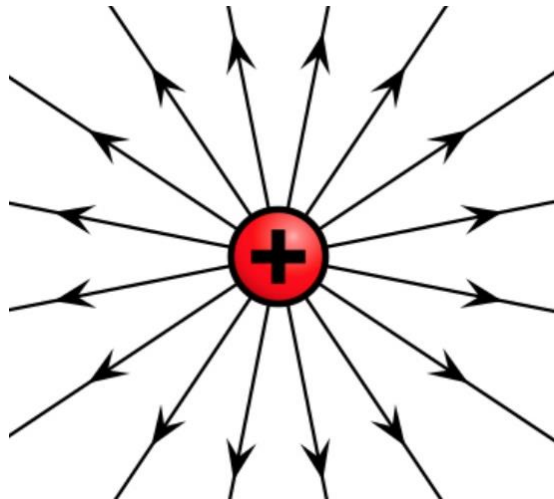


UMS log in >> UMS navigation >> Learning Management System >> Academic course syllabus >> syllabus files >> select the session >> PHY110

Syllabus

Unit I

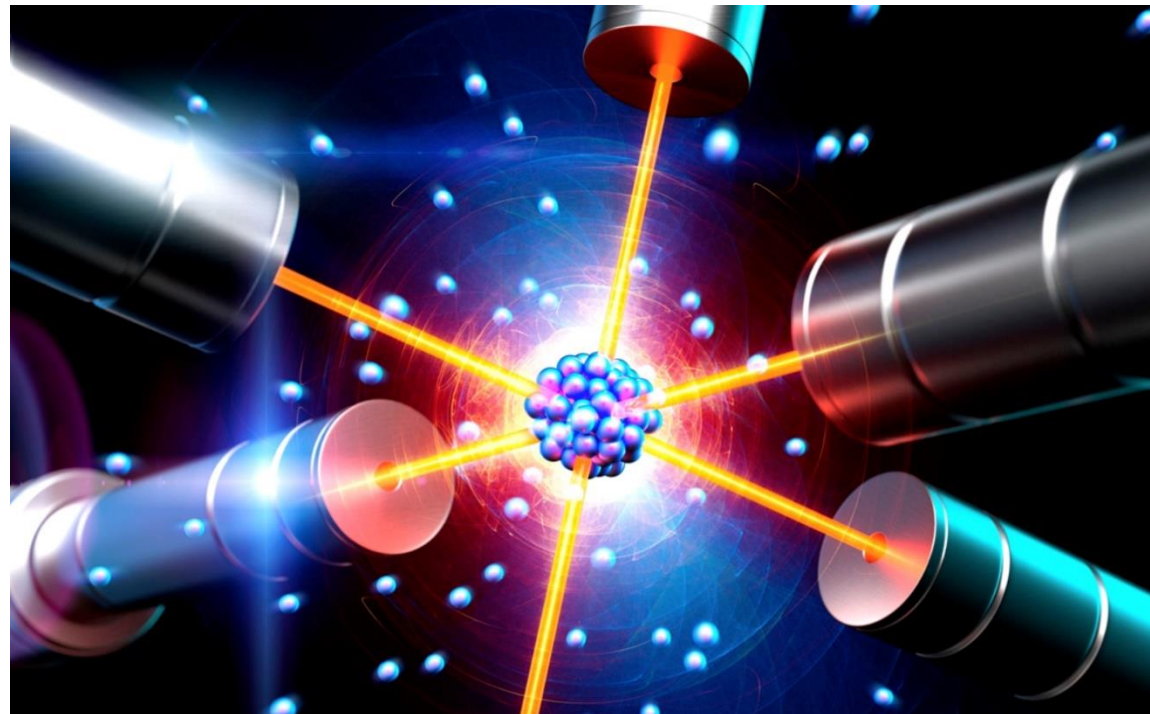
Electromagnetic theory : scalar and vectors fields, concept of gradient, divergence and curl, Gauss theorem and Stokes theorem (qualitative), Poisson and Laplace equations, continuity equation, Maxwell electromagnetic equations (differential and integral forms), physical significance of Maxwell equations, Ampere Circuital Law, Maxwell displacement current and correction in Ampere Circuital Law



Unit II

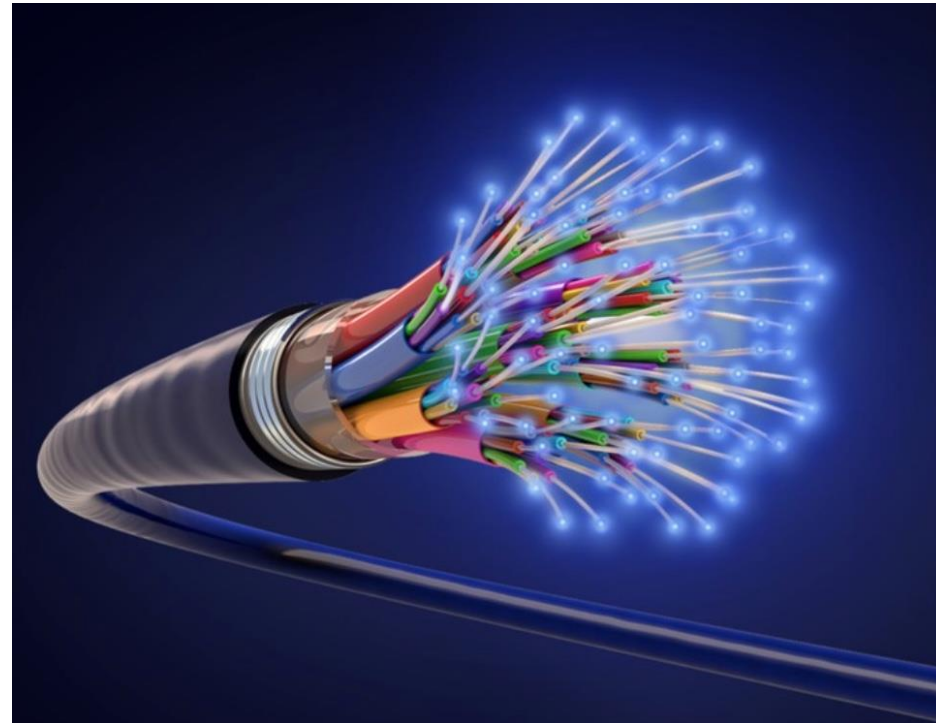


Lasers and applications : fundamentals of laser- energy levels in atoms, Radiation matter interaction, Absorption of light, spontaneous emission of light, stimulated emission of light, population of energy levels, Einstein A and B coefficients, metastable state, population inversion, resonant cavity, excitation mechanisms, Nd - YAG, He-Ne Laser, Semiconductor Laser, lasing action, properties of laser, applications of laser: holography



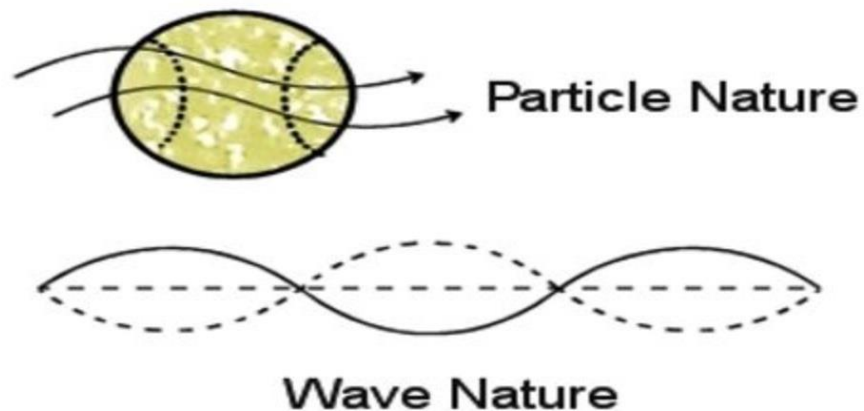
Unit III

Fiber optics : fiber optics introduction, optical fiber as a dielectric wave guide, total internal reflection, acceptance angle, numerical aperture, relative refractive index, V-number, step index and graded index fibers, losses associated with optical fibers



Unit IV

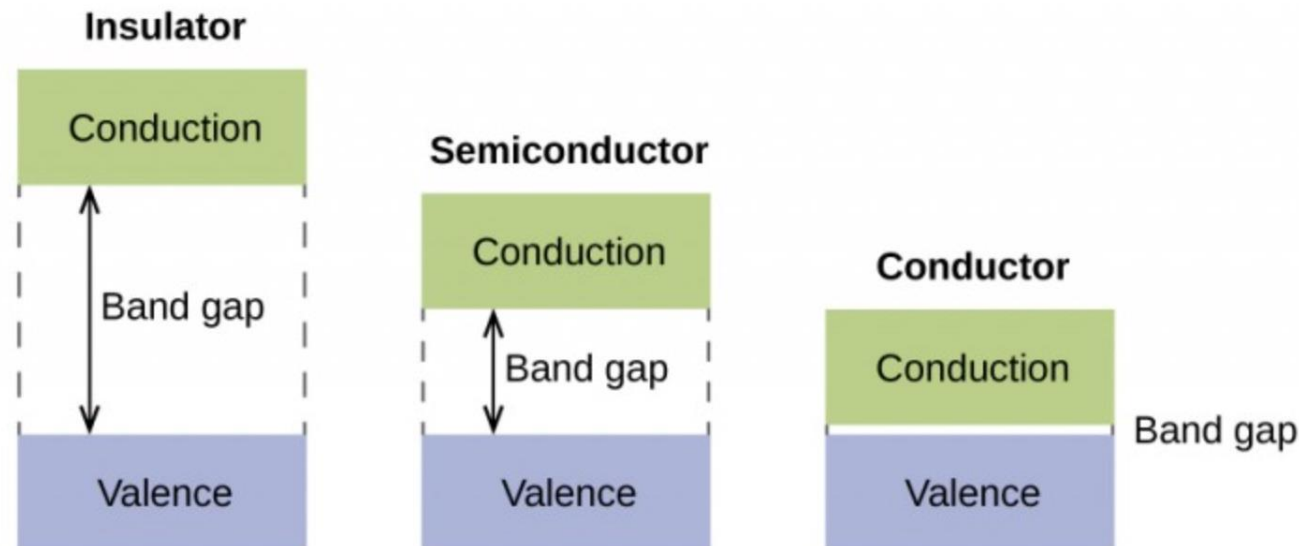
Quantum mechanics : need of quantum mechanics, photoelectric effect, concept of de Broglie matter waves, wavelength of matter waves in different forms, Heisenberg uncertainty principle, concept of phase velocity and group velocity (qualitative), wave function and its significance, Schrodinger time dependent and independent equation, particle in a box, tunneling effect (Qualitative idea)



$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

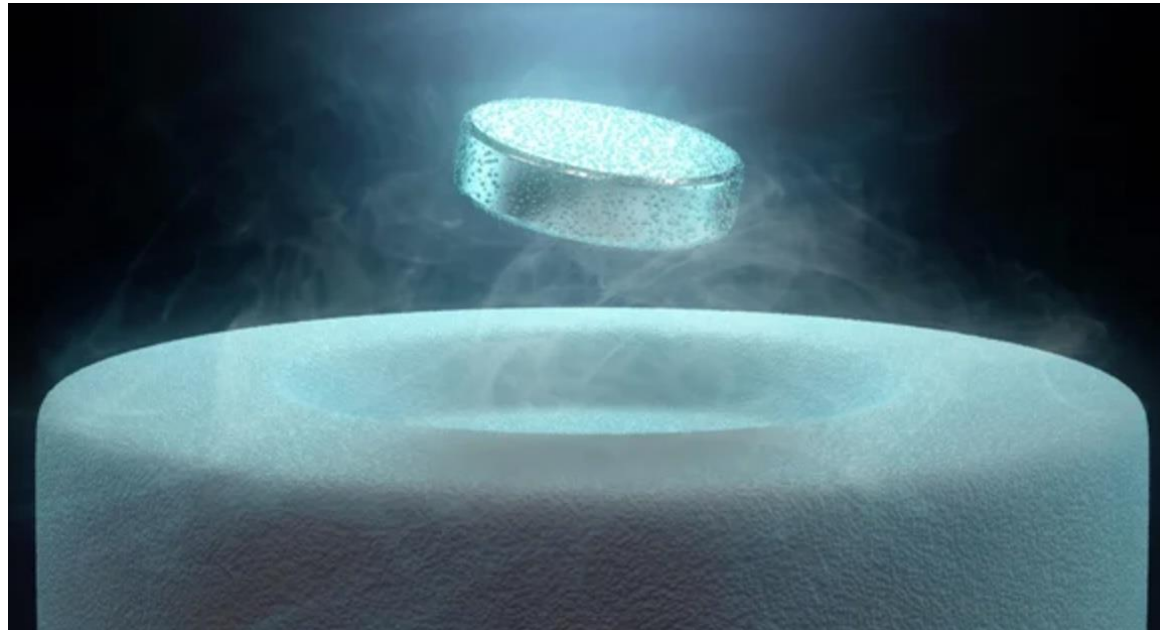
Unit V

Solid state physics : free electron theory (Introduction), diffusion and drift current (qualitative), fermi energy, fermi-dirac distribution function, and theory of solids -formation of allowed and forbidden energy bands, concept of effective mass - electrons and holes, Hall effect (with derivation), semiconductors and insulators, fermi level for intrinsic and extrinsic semiconductors, direct and indirect band gap semiconductors, solar cell basics



Unit VI

Introduction to engineering materials : dielectric materials definition, dielectric constant, magnetic materials: dia, para, ferromagnetic materials, magnetic data storage, piezoelectric materials: direct and inverse piezoelectric methods, superconducting materials: properties, Meissner effect, Type I & Type II superconductors, applications



Text books and Reference Books

Text Books:

1. ENGINEERING PHYSICS by B K PANDEY, S. CHATURVEDI, CENGAGE LEARNING

References:

1. ENGINEERING PHYSICS by HITENDRA K MALIK, A K SINGH, Tata McGraw Hill, India
2. CONCEPT OF MODERN PHYSICS. by BESIER ARTHUR., MCGRAW HILL EDUCATION
3. FUNDAMENTAL OF PHYSICS by HALLIDAY D., RESNICK R., WALKER J., WILEY

Course Orientation

1. COMPETITIVE EXAMINATION (Higher Education)
2. COMPETITIVE EXAMINATION (Civil Services)
3. KNOWLEDGE ENHANCEMENT

Assessment Model

L: 3.0 T: 0.0 P: 0.0 Cr: 3.0

Weightages:

Attendance: 5

CA: 25 (total CA - 3, 1 compulsory, best 1 in remaining 2)

MTT: 20 (Unit I,II and III)

ETT: 50 (Unit I to VI)

Weightages:

1. Attendance:

Criteria	Obtained	Marks
(a) If your attendance is between 90-100 %	5	
(b) If your attendance is between 85-90%	4	
(c) If your attendance is between 80-85%	3	
(d) If your attendance is between 75-80%	2	

Minimum attendance required to sit in ETT is 75%

*** Note: No converge factor is applicable: for e.g. if your attendance is 74.99 it will not be consider as 75**

Weightages:

2. Continuous Assessments (CA): 25

Type	Total Marks
(a) Two Subjective CA (Consider 1 best)	30 marks each
(b) One Assignment based CA (Compulsory)	30 marks

Check your understanding ! (Q and A with the students)

